

Project Management Plan



CRISTAR

University of Toronto

**Submitted in conformity with the requirements
for the CAN-RGX IX program**

Submitted: December 30, 2025

Executive Summary:

The CRISTAR project aims to study laser-induced crystallization under microgravity conditions as part of the CAN-RGX IX program. Crystallization is a critical process across multiple technological fields, and recent advances in laser-induced crystallization have enabled rapid, highly controlled crystal formation on Earth; however, the role of gravity in these processes remains poorly understood. By comparing the morphology and structural characteristics of crystals formed using laser-induced crystallization in a microgravity environment to those formed using the same technique on Earth, this project aims to advance the development of space-based crystallization techniques.

The CRISTAR team is composed of five undergraduate engineering students, two graduate physics students, and a supporting faculty advisor, all from the University of Toronto. To manage the technical requirements and timeline, the project is organized into five phases: design development, ground testing, pre-flight campaign, flight campaign, and post-flight review. Each phase includes clearly defined objectives, deliverables, and responsibilities, supported by a detailed work breakdown structure, timeline, and contingency protocols. Flight safety, laser safety, and experimental reliability are prioritized throughout the project lifecycle. A formal task reallocation and member recruitment process has been established to mitigate risks associated with team member availability and workload changes.

Budget requirements total \$7,850 CAD, with a conservative fundraising target of \$11,775 CAD after applying a 1.5x safety factor. Funding is pursued through a combination of “in-kind” donations, university funding programs, student organizational funding, and specialized industry support, with each member being responsible to secure funding for their area of expertise.

In parallel with technical objectives, the project emphasizes public engagement through ongoing in-person and digital outreach activities. These efforts aim to communicate the scientific and societal value of microgravity research to students and the broader community.

Contents:

Executive Summary:.....	2
Introduction: Project Background	4
Team structure and Management	5
Team Members.....	6
Task Reallocation Protocol.....	7
Member Recruitment.....	8
Project Timeline	9
Time Management Protocol	9
Project Time Allocation.....	9
Budget and Funding.....	13
Budget and funding strategy	Error! Bookmark not defined.
References.....	16

List of Figures:

Figure 1: Work breakdown structure for project tasks and responsibilities for each member.....	5
Figure 2: Step by step plan of action for task reallocation after a team member requests a reduction in contribution expectations or is unable to continue their support of the CRISTAR project.....	7
Figure 3:Step by step plan for the recruitment of team members which will be applied when new team members are required to support project goals.....	8

List of Tables:

Table 1: List of team members, including the faculty advisor, all from the University of Toronto, with roles identified.	6
Table 2: Overview of the time allocations for each phase of the project.	10
Table 3: Time and task allocation for the design development phase.....	11
Table 4: Time and task allocation for the ground testing phase.....	12
Table 5: Time and task allocation for the flight campaign phase.	12
Table 6: Time and task allocation for the post flight phase of the project.	12
Table 7:Budget requirements for project components are travel costs are estimated and described. Shipping costs are planned to be covered by the Miller Lab and are not included in the budget below....	14
Table 8: Funding Sources and Member Responsibilities: Each team member is accountable for securing resources related to their assigned engineering tasks. Responsibilities include identifying required components and contacting suppliers to explore potential donations or discounts for those items.....	15

Introduction: Project Background

Crystallization is a foundational process in materials science, with applications in pharmaceutical drug development¹⁻⁵, organic electronics⁶⁻⁸, energy storage^{9,10}, and semiconductor manufacturing¹¹. Interest in space-based crystallization processes is on the rise and has developed into a distinct part of the space industry¹²⁻¹⁶. A promising new ground-based technique involves laser-induced crystallization¹⁷, where short-lived bubbles rapidly reduce pressure and trigger nucleation. This method shortens crystal formation time from hours to seconds, with enhanced control over crystal size¹⁸, offering potential for high-throughput production of nano- and microcrystals¹⁸. However, the role of gravity in these dynamics, especially in laser-induced methods, remains poorly understood. Clarifying this influence is important for advancing space-based manufacturing and improving Earth-based simulations of space-grown materials.

This research aims to demonstrate laser-based crystallization in a microgravity environment and to compare the resulting crystal structures with those formed on Earth. The experiment will assess whether laser-induced crystallization under microgravity conditions accelerates crystal formation and enhances sample quality relative to Earth-grown crystals. These insights are particularly relevant for producing crystals suitable for diffraction-based structural determination using electron beams, which are now sufficiently compact and portable for spaceflight applications.

Key stakeholders include pharmaceutical companies that depend on high-quality crystals for drug development¹⁻⁵, Canadian startups looking for new niches in the space industry¹⁹, and the Canadian Space Agency, which supports research that strengthens Canada's role in the emerging space economy. Environmental considerations are also important. The proposed laser-based crystallization approach requires only short suborbital flights to produce thousands of samples, rather than relying on traditional orbital missions. Laser induced crystallization would reduce the number of satellites in low Earth orbit, mitigating space debris, require less fuel per flight, lower the overall environmental impact of space-based crystallization processes, and decrease operational costs.

The motivation behind this experiment is to lay the groundwork for scalable crystallization techniques in space, enabling space-based manufacturing, high-throughput screening and structural analysis using electron diffraction. By bridging fundamental science with engineering implementation, this research addresses critical questions about gravity's role in materials science.

The goal of the proposed experiment is to demonstrate the production of microcrystals using laser cavitation-based crystallization in a short-duration microgravity environment, and to compare them to crystals produced using the same technique under Earth's gravity. This objective includes demonstrating that laser-induced crystallization can reliably occur in microgravity and determining whether gravity influences the morphology and structural characteristics of the resulting crystals.

Team structure and Management

CRISTAR is a team of seven students at the university of Toronto, with one faculty advisor. The team consists of two graduate students from the department of physics, and five undergraduate students from the Department of Engineering. To ensure the successful completion of our project objectives, task have been distributed as outlined in the work breakdown structure in Figure 1, with role assignment being given in Table 1. Roles were allocated according to areas of expertise and interest. We decided to have concrete roles to make job alignment and accountability straightforward. However, everyone is expected to help outside of their role when needed since interdisciplinary collaboration is at the centre of the team ethos.

The work breakdown structure in Figure 1 is written in terms of lines of effort instead of the five project phases to better illustrate the team workload distribution.

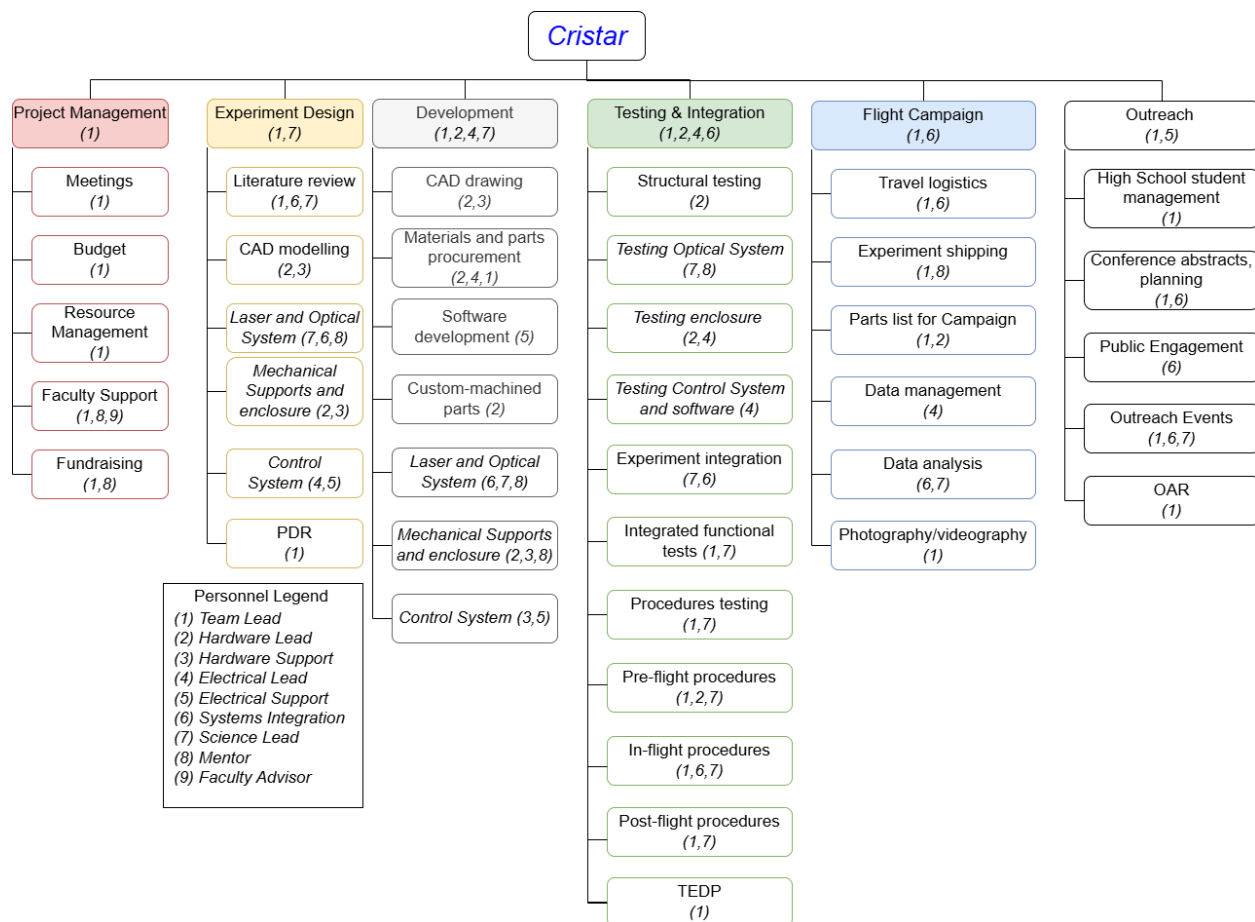


Figure 1: Work breakdown structure for project tasks and responsibilities for each member.

Team Members

Table 1: List of team members, including the faculty advisor, all from the University of Toronto, with roles identified.

First, Last Name	Degree in progress	Level of Study	Department	Role	Email
Alexander, Wainwright	Ph.D.	Graduate	Physics	Team Lead	Alexander.Wainwright@mail.utoronto.ca
Syeda, Mahdia	BASc.	Undergraduate	Engineering Sciences	Science Lead	Syeda.Mahdia @mail.utoronto.ca
Lauren, Altomare	BASc.	Undergraduate	Engineering Sciences	Systems Integration	Lauren.Altomare@mail.utoronto.ca
Daniel, Yu	BASc.	Undergraduate	Computer Engineering	Electrical and Hardware Support	Danielz.Yu@mail.utoronto.ca
Elias, Barsa	BASc.	Undergraduate	Engineering Sciences	Hardware Lead	Elias.Barsa@mail.utoronto.ca
Gabriel, Caribe	BASc.	Undergraduate	Engineering Sciences	Electrical Lead	G.Caribe@mail.utoronto.ca
Khaled, Madhoun	Ph.D.	Graduate	Physics	Mentor	Khaled.Madhoun@mail.utoronto.ca
R.J. Dwayne, Miller	-	University Professor	Physics & Chemistry	Faculty Advisor	Dwayne.Miller@utoronto.ca

Task Reallocation Protocol

Contingency planning is key to project success. As life changes may prevent CRISTAR members from continuing their commitment to the team, we have developed a task reallocation protocol, shown in Figure 2, to be followed if team members require a reduction in tasks or are unable to continue their commitment to the project.

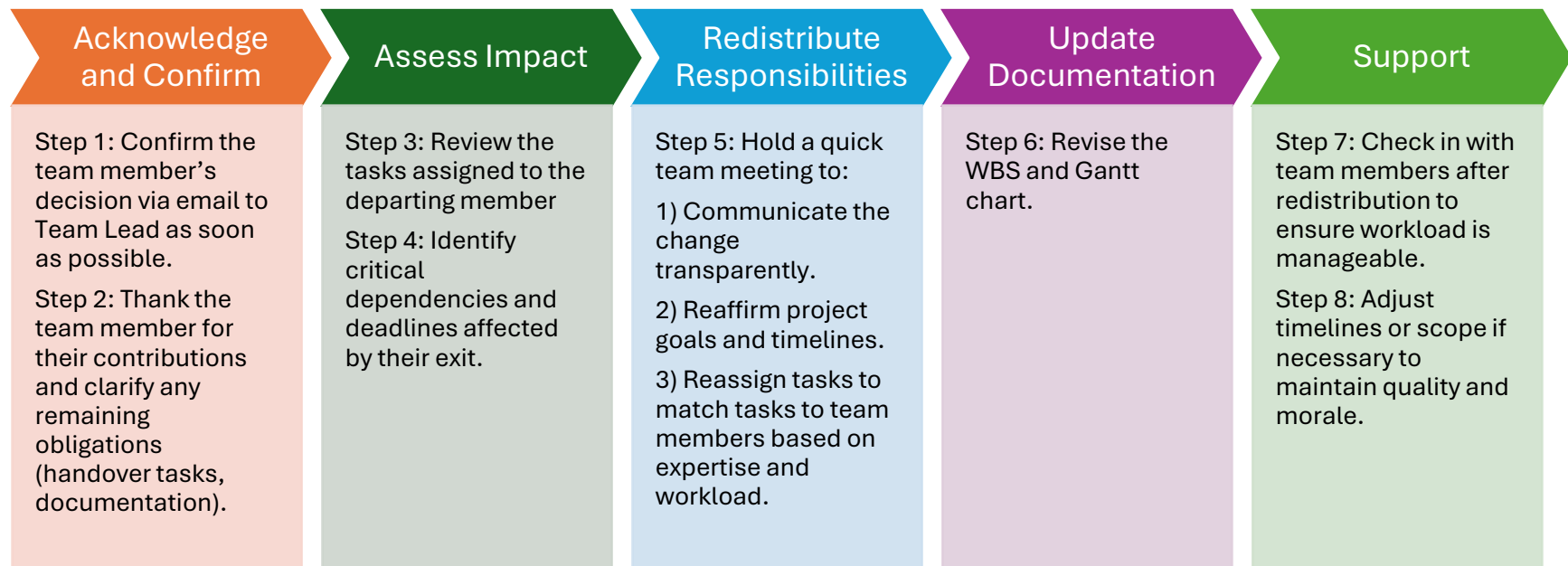


Figure 2: Step by step plan of action for task reallocation after a team member requests a reduction in contribution expectations or is unable to continue their support of the CRISTAR project.

Member Recruitment

As the project develops, there may be a need to recruit new members. In such situations, the steps outlined in Figure 3 will be used to find a suitable candidate. Currently, we are reviewing applications for a social media position on the team. This individual will provide the online media presence for CRISTAR and help run outreach activities.



Figure 3: Step by step plan for the recruitment of team members which will be applied when new team members are required to support project goals.

Project Timeline

Time Management Protocol

The team will maintain schedule adherence through structured planning and proactive monitoring. A detailed project timeline with defined milestones has been established, supported by a [Gantt chart](#) for real-time visibility (examples seen in Tables 2-6 below). Weekly team meetings, led by the Team Leader, will begin with a progress review, highlighting completed tasks and upcoming milestones to ensure alignment with the project plan. The Gantt chart will be reviewed and updated during these meetings to provide a clear visual reminder of progress and deadlines. Each design leader will be responsible for identifying which tasks are essential to meeting project objectives versus extra or optional tasks, allowing the team to make informed decisions on where time can be saved if necessary. Time allocation will follow a strict priority order: flight safety first, followed by scientific considerations, then media and outreach activities. Contingency buffers will be integrated into critical path activities to mitigate potential risks, and any emerging issues will be addressed promptly during these reviews.

To reinforce accountability and efficiency, the Team Leader will coordinate closely with all members to ensure tasks are completed on time and resources are optimally allocated. Clear communication channels (Google Drive for file transfer, and communication via the team WhatsApp and Discord channel) will be maintained to provide timely updates and resolve bottlenecks before they impact the schedule. Each team member will be responsible for accurate progress reporting and collaborating effectively to meet deadlines, while the Team Leader will oversee adherence to the timeline and implement minor adjustments as needed to keep the project on track.

Project Time Allocation

Given the short timeline and demanding engineering requirements, the project has been divided into five distinct phases: design development, ground testing, pre-flight campaign, flight campaign, and post-flight review. During the design development phase, we will create detailed concepts for each component of the microgravity system. Each component will first undergo independent testing to ensure compliance with program requirements and scientific objectives. Once verified, the components will be integrated for ground testing, where we will assess overall system performance against engineering standards for flight safety, laser safety, and experimental reliability. Following ground testing, the pre-flight campaign phase will simulate in-flight procedures on the ground to validate operational readiness. The system will then be deployed for microgravity flights and experiment execution. Finally, in the post-flight review phase, we will analyze results, publish findings, and support ongoing public outreach. In parallel with these engineering objectives, we will conduct both in-person and digital outreach programs to engage the broader community. Details as to task and time allocations for the project are given in Tables 2-6 below.

Table 2: Overview of the time allocations for each phase of the project.

TASK	ASSIGNED	PROGRESS		START	DAYS	END
Design Development				10/12/2025		3/2/2026
Ground Testing				10/1/2026		13/4/2026
Pre-Flight Testing				23/2/2026		3/6/2026
Flight Test				22/6/2026		30/6/2026
Post Fligth Review				30/6/2026		20/8/2026

Table 3: Time and task allocation for the design development phase.

TASK	ASSIGNED	START	END
Design Development		10/12/2025	3/2/2026
Structure and Organization Development	Alex	10/12/2025	15/12/2025
Finish Verification and Compliance Matrix	Alex	15/12/2025	22/12/2025
Preliminary Design Development (Optical system)	Syeda & Khaled	22/12/2025	5/1/2026
Preliminary Design Development (Science)	Alex	22/12/2025	5/1/2026
Preliminary Design Development (Electrical)	Gabriel	22/12/2025	5/1/2026
Preliminary Design Development (Hardware)	Elias	22/12/2025	5/1/2026
Preliminary Design evaluation and selection (Optical system)	Syeda & Khaled	5/1/2026	6/1/2026
Preliminary Design evaluation and selection (Samples)	Lauren	5/1/2026	6/1/2026
Preliminary Design evaluation and selection (Electrical)	Gabriel & Daniel	5/1/2026	6/1/2026
Preliminary Design evaluation and selection (Mechanical)	Elias & Daniel	5/1/2026	6/1/2026
Flight Procedure Development	Syeda & Lauren	10/12/2025	5/1/2026
Ground Test Plan Development	Elias & Gabriel	5/1/2026	3/2/2026
Check design with SMEs (Particular attention to fasteners, loads, and laser safety)	Alex	6/1/2026	12/1/2026
Preliminary Design stress test on CAD (Mechanical)	Elias	5/1/2026	11/1/2026
Budget and Fundraising Finalization	Alex	6/1/2026	12/1/2026
Outreach Plan Finalization	Alex	10/12/2025	10/1/2026
PDR 1 and Test Equipment Data Package Rev 1	Alex	6/1/2026	10/1/2026

Table 4: Time and task allocation for the ground testing phase.

TASK	ASSIGNED	START	End
Ground Testing		10/1/2026	13/4/2026
Order parts	Alex	6/1/2026	7/1/2026
Optical System and interlock Prototype	Khaled	6/1/2026	19/1/2026
Prototype Crystal generation	Alex	6/1/2026	4/2/2026
Prototype electrical design	Gabriel	6/1/2026	4/2/2026
Prototype Mechanical design	Elias & Daniel	6/1/2026	4/2/2026
Stress test Mechanical Design	Elias	4/2/2026	10/2/2026
Stress test integrated Electrical Design	Khaled	4/2/2026	10/2/2026
Preliminary Design Review	Alex	20/2/2026	23/2/2026
Integrate components	Lauren	10/2/2026	16/2/2026
Stress test total system	Elias	16/2/2026	1/3/2026
PDR 1 and Test Equipment Data Package Rev 2	Alex	10/4/2026	13/4/2026

Table 5: Time and task allocation for the flight campaign phase.

TASK	ASSIGNED	START	End
Flight Campaign		22/6/2026	30/6/2026
Flights	Alex	22/6/2026	30/6/2026
Electron Diffraction	Syeda	30/6/2026	29/7/2026

Table 6: Time and task allocation for the post flight phase of the project.

TASK	ASSIGNED	START	End
Post Flight Review		30/6/2026	20/8/2026
Post Flight Survey	Alex	30/6/2026	22/7/2026
Outreach Activity Report	Alex	30/6/2026	7/8/2026
Manuscript writing	Alex	22/7/2026	20/8/2026

Budget and Funding

Major budget requirements for the project include material costs for essential components, travel expenses, accommodation for the flight campaign, and shipping costs for transporting project components to and from the National Research Council (NRC). These expenses have been estimated and summarized in Table 2, with the total project cost projected at \$7,850 CAD. To ensure sufficient funds are available, a factor of safety of 1.5 has been applied, resulting in a fundraising goal of \$11,775 CAD. To date, significant support has been received from the Miller Lab at the University of Toronto and the Laser Safety Team, reducing costs by \$1,050 CAD. Consequently, the current fundraising goal stands at \$10,200 CAD.

Our funding strategy focuses on three key approaches: Connecting, Applying, and Collaborating. First, we aim to connect with corporations to secure “in-kind” support by requesting optical / mechanical components and seeking discounts where donations are not possible. Second, we will apply for financial assistance through university funding programs and student organizational funding opportunities to ensure adequate resources for the project. Finally, we plan to collaborate with industry stakeholders by requesting specialized support, such as partnering with Hitachi for nano-crystallography work. These combined efforts will help us secure both material and financial resources while fostering strategic partnerships. An overview of our funding requests are in Table 8. At the time of writing only two university funding programs have been secured. The most time sensitive funding sources currently are the “in-kind” requests to component manufacturers: Thorlabs, Zaber, McMaster-Carr, and Kentek. These requests have not been sent yet at time of writing, but the relevant team member will reach out to each company in conjunction with ordering the required parts by the January 7th deadline. The aim is to send out all further funding requests to the university and collaborating companies by the end of January.

Table 7: Budget requirements for project components are travel costs are estimated and described. Shipping costs are planned to be covered by the Miller Lab and are not included in the budget below.

Item	Description	Estimated Cost (CAD)
Translation Stages	Stages to move the crystallization sample between trials.	\$3000.00
Camera	Camera to image the cavitation process in the crystallization solution	\$500.00
Focusing System	Lens or mirror to focus the laser light inside the crystallization solution to drive cavitation.	\$2000.00
Camera Mount	Mount to hold the camera security and withstand the strong acceleration forces present during parables.	\$300.00
Interlock system	System to shut down the operations of the laser if the laser chamber is opened.	\$50.00
Enclosure materials	Laser safe materials required to enclose the cavitation chamber.	\$500.00
Metal for Fixed Optics	Materials to fix and hold optics in place in the high acceleration environment of the experiment.	\$500.00
Travel Costs	Travel for one week in the Ottawa area for the flight campaign phase of the project.	\$1000.00
Total Cost		\$7850.00

Table 8: Funding Sources and Member Responsibilities: Each team member is accountable for securing resources related to their assigned engineering tasks. Responsibilities include identifying required components and contacting suppliers to explore potential donations or discounts for those items.

Funding Source	Member Responsible	Confirmed (Y/N)	Amount Expected (CAD)	Amount Received (CAD)
SPIE Student Chapter	Alex	Yes	\$1000	\$1000.00
Humans Learning Machine Learning	Alex	Yes	\$200	\$200.00
Massey College	Alex	No	\$100	\$0.00
Physics Department	Khaled	No	\$100	\$0.00
Department of Chemistry	Khaled	No	\$100	\$0.00
Thorlabs	Gabriel	No	\$3000	\$0.00
Zaber	Syeda	No	\$300	\$0.00
McMaster-Carr	Elias	No	\$1000	\$0.00
Kentek	Syeda	No	\$500	\$0.00
EngSoc CPSIF Funding	Lauren	No	\$200	\$0.00
Student Initiative Funding	Lauren	No	\$500	\$0.00
Light Matter Interaction Inc.	Alex	No	\$200	\$0.00
Hitachi	Faculty Advisor	No	\$1000	\$0.00
Total Funding			\$10900	\$1200.00

References

1. Reichert, P. *et al.* Pembrolizumab microgravity crystallization experimentation. *npj Microgravity* **5**, 28 (2019).
2. McPherson, A. & DeLucas, L. J. Microgravity protein crystallization. *NPJ Microgravity* **1**, 15010 (2015).
3. Orehek, J., Teslić, D. & Likozar, B. Continuous Crystallization Processes in Pharmaceutical Manufacturing: A Review. *Org. Process Res. Dev.* **25**, 16–42 (2021).
4. Mukherjee, S. Chapter 2 - Isolation and Purification of Industrial Enzymes: Advances in Enzyme Technology. in *Advances in Enzyme Technology* (eds. Singh, R. S., Singhanian, R. R., Pandey, A. & Larroche, C.) 41–70 (Elsevier, 2019). doi:10.1016/B978-0-444-64114-4.00002-9.
5. Crystallizing Proteins in Space Helping to Identify Potential Treatments for Diseases - NASA. <https://www.nasa.gov/missions/station/iss-research/crystallizing-proteins-in-space-helping-to-identify-potential-treatments-for-diseases/> (2022).
6. Peng, B., Wu, R. & Li, H. Crystallization from a Droplet: Single-Crystalline Arrays and Heterojunctions for Organic Electronics. *Acc. Chem. Res.* **54**, 4498–4507 (2021).
7. Yu, P., Zhen, Y., Dong, H. & Hu, W. Crystal Engineering of Organic Optoelectronic Materials. *Chem* **5**, 2814–2853 (2019).
8. Johnson, H. M. *et al.* Discovery of Crystallizable Organic Semiconductors with Machine Learning. *J. Am. Chem. Soc.* **146**, 21583–21590 (2024).
9. Dong, W. & Huang, F. Understanding the influence of crystal packing density on electrochemical energy storage materials. *eScience* **4**, 100158 (2024).
10. Zhang, R. *et al.* Balancing volumetric and gravimetric capacity for hydrogen in supramolecular crystals. *Nat. Chem.* **16**, 1982–1988 (2024).
11. Wilkinson, A. R. *et al.* A meta-analysis of semiconductor materials fabricated in microgravity. *npj Microgravity* **10**, 73 (2024).
12. Redwire to Develop Space-Based Drug Development and Manufacturing Platform. *Redwire Space* <https://rdw.com/newsroom/redwire-to-develop-space-based-drug-development-and-manufacturing-platform/> (2022).
13. Crystal Growth. *ISS National Lab* <https://issnationallab.org/research-and-science/space-research-overview/research-areas/in-space-production-applications/crystal-growth/>.

14. Differentiated drug products enabled by microgravity • Varda Space Industries. *Varda Space Industries*
<https://www.varda.com/biopharma/>.
15. Varda raises \$187M to scale in-space drug crystallization lab | Pharma Manufacturing.
<https://www.pharmamanufacturing.com/industry-news/news/55302723/vara-raises-187m-to-scale-in-space-drug-crystallization-lab>.
16. Mulligan, M. K., Tuma, S., Mullins, S., Savin, K. A. & Wilson, A. Protein Crystallization in Microgravity: Commercialization and the Next Chapter. *Curr Stem Cell Rep* **11**, 6 (2025).
17. Korede, V. *et al.* A Review of Laser-Induced Crystallization from Solution. *Crystal Growth & Design* **23**, 3873–3916 (2023).
18. Nikbin, E. *et al.* Femtosecond Laser-Induced Crystallization of Cytosine Monohydrate for Serial Nano Electron Diffraction Analysis. *Microscopy and Microanalysis* vol. 31 (2025).
19. Mortillaro, N. Canadian companies race to launch rockets into orbit from home soil — a first. *CBC News* (2025).
20. Wright, H. *et al.* An Analysis of Publicly Available Microgravity Crystallization Data: Emergent Themes Across Crystal Types. *Crystal Growth & Design* **22**, 6849–6851 (2022).
21. Pal, K. & Radocea, A. Gravity as a Knob for Tuning Particle Size Distributions of Small Molecules. *Crystal Growth & Design* **24**, 2370–2383 (2024).
22. Timofeev, V. I., Ivanovsky, A. S., Marchenkova, M. A., Rakitina, T. V. & Kuranova, I. P. Growth of protein crystals in microgravity: history and present status. *Crystallography Reviews* **30**, 236–260 (2024).
23. Paulson, J. *et al.* Microgravity-Grown Crystals as Seeds for Pharmaceutical Compounds. *Crystals* **15**, 825 (2025).
24. Takahashi, S. *et al.* JAXA protein crystallization in space: ongoing improvements for growing high-quality crystals. *J Synchrotron Radiat* **20**, 968–973 (2013).